

MAXplain: A Multi-Agent System for Interactive Multimodal Hate Speech Detection

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Abstract

Multimodal hate speech detection targets offensive content expressed through combinations of modalities such as text and images, which often evade detection when analyzed separately. We introduce MAXplain, an interactive framework that addresses both issues via a configurable LLM-based multi-agent architecture. Specialized agents handle distinct subtasks and exchange information through structured dialogues, enabling intrinsic explainability and improved accuracy. The web interface supports human-in-the-loop interaction, including real-time adjustment of agent behaviors and evaluation rules. A browser plugin enables direct inspection of online content. While demonstrated for hate speech detection, MAXplain also supports rapid prototyping for other multimodal tasks.

CCS Concepts

• **Computing methodologies** → **Multi-agent systems**; *Visual content-based indexing and retrieval*; • **Human-centered computing** → *HCI design and evaluation methods*.

Keywords

Multi-agent systems, Explainable AI, Large language models, Multimodal analysis, Hateful meme detection, Human-in-the-loop

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1 Introduction

Hate speech detection (HSD) methods are becoming increasingly relevant as communication continues to shift into online spaces. Compared with text-only content, multimodal hate speech significantly challenges existing unimodal solutions [3, 8] due to the need

for understanding the full context via interactions between modalities, such as images and texts in captioned images and memes. Furthermore, a lack of explainability and human involvement in current systems hampers trustworthy moderation, prevents the incorporation of domain expertise, and raises concerns about fairness and accountability [5, 9].

Although large-scale sub-symbolic models such as LLMs offer strong performance, their decision-making is often difficult to interpret due to the lack of explicit reasoning structures, which represents a fundamental challenge in explainable AI [2]. Post-hoc methods such as SHAP or LIME provide only localized explanations for individual predictions and fail to convey a systematic understanding of global behavior [2, 7]. Also, in the multimodal context, explanations should be more nuanced and modality-aware. Rudin [11] argues that in safety-critical contexts, intrinsically interpretable models should be preferred over opaque black-box systems, which is particularly relevant for hate speech moderation.

We therefore present the Multi-Agent Explainability Framework (MAXplain), an interactive system comprising collaborative agents that provide inherent explainability via structured dialogues. Each agent is responsible for a specific subtask, such as analyzing text, images, or contextual clues. The agents independently process their input, exchange intermediate results, and collectively contribute to a transparent chain of reasoning. A final designated agent then integrates these results to make the overall classification decision. The system enables human-in-the-loop interaction and can be flexibly adapted to various use cases, including parental control over social media content, verification of user-reported posts, or accelerating the annotation of training data. MAXplain thus increases transparency, trust, and efficiency specifically for moderating multimodal hate speech. A project demonstration will be provided at <https://maxplain.com>.

2 System Architecture

MAXplain uses a layered architecture with distinct backend and frontend components, as shown in Figure 1. The backend is a SPADE-based Multi-Agent System (MAS) enabling collaboration between specialized LLM-based agents to process multimodal content. The frontend provides two access points: a Gradio-based [1] web interface for system management, and a lightweight Chrome browser plugin for extracting media content from online platforms.

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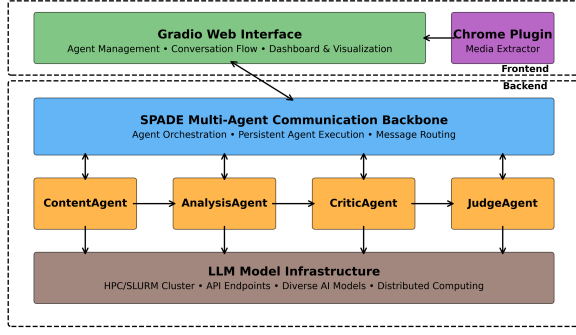


Figure 1: MAXplain system architecture.

2.1 Backend: MAS Communication Backbone

The communication infrastructure comprises decentralized SPADE-based [10] agents that interact over the XMPP communication protocol. Each agent created through the web interface is registered as a dedicated XMPP account and deployed on a remote high-performance computing environment (GPU-based SLURM cluster with multi-node parallel execution). Once initialized, agents remain idle and listen for incoming messages, eliminating startup delays and enabling token-level streaming for real-time output. Alternatively, agents can run locally, using models accessed through API endpoints to support flexible deployment.

Agent Roles: In an example task decomposition setup (see Figure 1), the ContentAgent describes visual and textual elements, the AnalysisAgent evaluates protected groups and attack severity, the CriticAgent challenges prior interpretations, and the JudgeAgent integrates all outputs for final classification. A central component is the configurable *rulebook*, encoding refined hate speech criteria such as contextual cues, protected attributes, cultural references, and platform-specific policies. The rulebook is accessible to all agents during runtime and guides their decisions.

Performance Evaluation: We evaluated our MAS on 1,000 balanced samples from the Hateful Memes dataset [8], comparing single-agent setups (with and without rulebook) to MAS setups using the least-to-most strategy [12]. This included a single-agent variant (all steps in one prompt) and a MAS setup with one agent per step. All configurations used Llama-3.2-11B-Vision-Instruct [6], except the best-performing one, where the JudgeAgent employed DeepSeek-R1-Distill-Qwen-32B (text-only) based solely on upstream outputs [4]. As shown in Table 1, the rulebook improved results. This illustrates how MAS architectures leverage specialized models for distinct subtasks in a single workflow.

2.2 Frontend: User Interfaces

Web-Based Management Interface: The central user interface enables comprehensive system management and interaction across three functional areas (see Figure 2): The Configuration Panel (Area A, orange) allows users to upload images, select datasets, configure agent roles and parameters, and define conversation flows including agent order, prompts, and rulebook integration. The Visualization

Table 1: Performance across different system configurations.

Configuration	Accuracy	F1 Score
Single Agent (no rulebook)	0.616	0.474
Single Agent (with rulebook)	0.644	0.536
Single Agent Task Decomposition	0.621	0.596
Multi-Agent (Judge: Llama-3.2-Vision)	0.666	0.698
Multi-Agent (Judge: DeepSeek-R1)	0.731	0.735

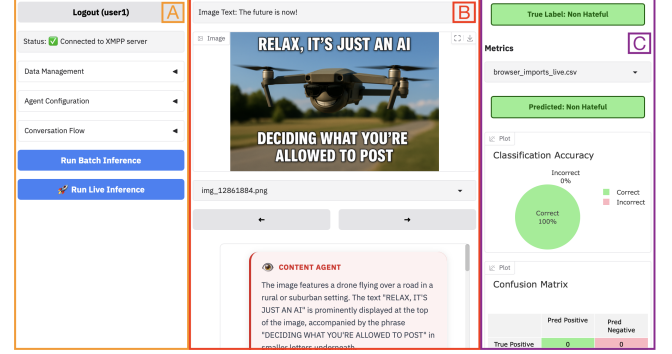


Figure 2: Screenshot of the MAXplain web interface.

Panel (Area B, red) displays image navigation and inter-agent communication steps. The Results Panel (Area C, purple) shows true and predicted labels, real-time inference progress, and key performance metrics such as accuracy, confusion matrix, and F1 score. These panels support iterative adjustment of agents, conversation logic, and evaluation rules in a human-in-the-loop workflow.

Media Extractor: A lightweight Chrome plugin uses a language model agent to extract content across platforms in a generalized manner. With one click, users can capture platforms and associated text (including post text, captions, or hashtags) directly from web-pages. The plugin integrates seamlessly with the backend, enabling immediate analysis without manual file transfers.

3 Conclusion

MAXplain improves the performance and explainability of automatic multimodal HSD by using an LLM-based MAS. Task decomposition is enabled by structured information exchange and the deployment of different models optimized for distinct subtasks. In our experiments on the Hateful Memes dataset, this architecture outperformed single-model baselines. The graphical interface allows real-time modification of agents, rules, and conversation flow without a restart. We have evaluated MAXplain for hate speech, but it can be adapted to various other tasks such as fake news detection with minimal adjustments. The framework supports rapid prototyping and the creation of proofs of concept. Future work includes fine-tuning each agent model for its dedicated role and incorporating additional modalities such as audio and video.

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